> **Complete proof library:** the full **M0–M6c corpus (1,406 pages)** is still available. Explore the seven chapters with exact page links · open the entire PDF directly · V0.6-specific proofs (24 pages).

READER'S EDITION · ENGLISH · WORKING PREPRINT 2026

Calendar-Time Maternal Genealogies

How shared ancestry changes generational dispersion

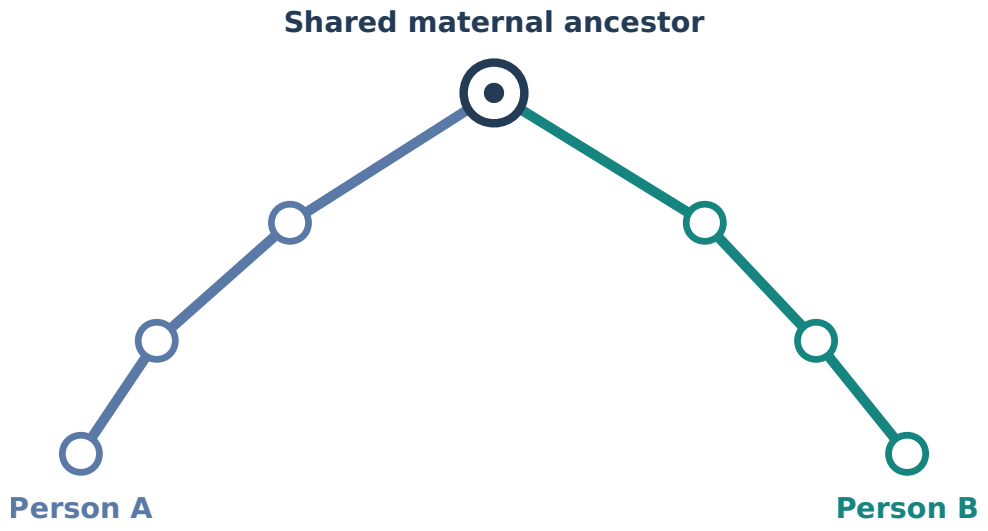
Léo Marquet · Bilingual reader's edition · Based on the mathematical manuscript V0.6

> Two people born today can have different numbers of maternal generations to the same historical date. And their lines may reach the very same ancestor.
> **What we prove is a mathematical mechanism, not a measurement of differences among real humans.**

1. The question behind the mathematics

Tracing a maternal lineage means following a person’s mother, her mother, and so on. Two people alive at the same time can count different numbers of generations to the same historical date because maternal ages at childbirth vary.

But genealogy is not a collection of independent paths. Two people can arrive at the **same female ancestor**. From that encounter backward, their older genealogical histories coincide: this is maternal **coalescence**.



Conceptual diagram: ancestral routes can differ until they meet; the earlier history is then shared. Not an observed pedigree.

Precise question. How much depth dispersion comes from variability in maternal ages, and how much depends on shared ancestral identities?

2. Two observables with different meanings

Write $\mathbf{G_i(t)}$ for the number of mother–child links individual *i* must cross to reach a common calendar boundary *t* time units in the past.

Mean pairwise gap

The average absolute difference in generation counts across all distinct pairs in a cohort.

Extreme range

The deepest maternal line's count minus the shallowest count in that cohort.

$$\Delta_N(t) = [1 \,/\, N(N-1)] \sum_{i \neq j} |G_i(t) - G_j(t)|$$
$$R_N(t) = \max_i G_i(t) - \min_i G_i(t)$$

Individual marginal distributions are not enough to recover collective dispersion. If all individuals receive the same depth on each realization, within-cohort disparity may be zero even though every person separately has substantial uncertainty. The *dependence structure* matters.

3. The explicit model actually proved

The technical manuscript first constructs a homogeneous mathematical population with $N_n = \lfloor \nu n \rfloor$ female identities at each birth date. Maternal ages take the abstract values 1 and 2 with probabilities p and 1−p, and each child selects a mother from the appropriate earlier birth-date pool.

$$P(A = 1) = p \; ; \; P(A = 2) = 1-p \; ; \; 0 < p < 1.$$

This model has variable generations and **genuine shared ancestral identities**. The units 1 and 2 are mathematical time units, *not* human reproductive ages.

Two-lineage limit (Theorem 4.1). The normalized difference in maternal depth has a Brownian limit stopped at the first shared ancestor. The limiting first-collision clock has an exponential rate $1/[\nu(2-p)^2]$. A finite collection of exogenous demographic regimes gives a Cox-process extension (Theorem 5.1).

> **Exact scope:** the proof concerns a homogeneous maternal pool and finitely many exogenous scenarios, not a complete reconstruction of historical populations with complex endogenous migration.

4. A theorem about a whole cohort

Exchangeability in the homogeneous model implies an exact finite-population identity: the expected cohort average over all distinct pairs equals the expectation for one uniformly selected pair. Independence between different pairs is *not* required.

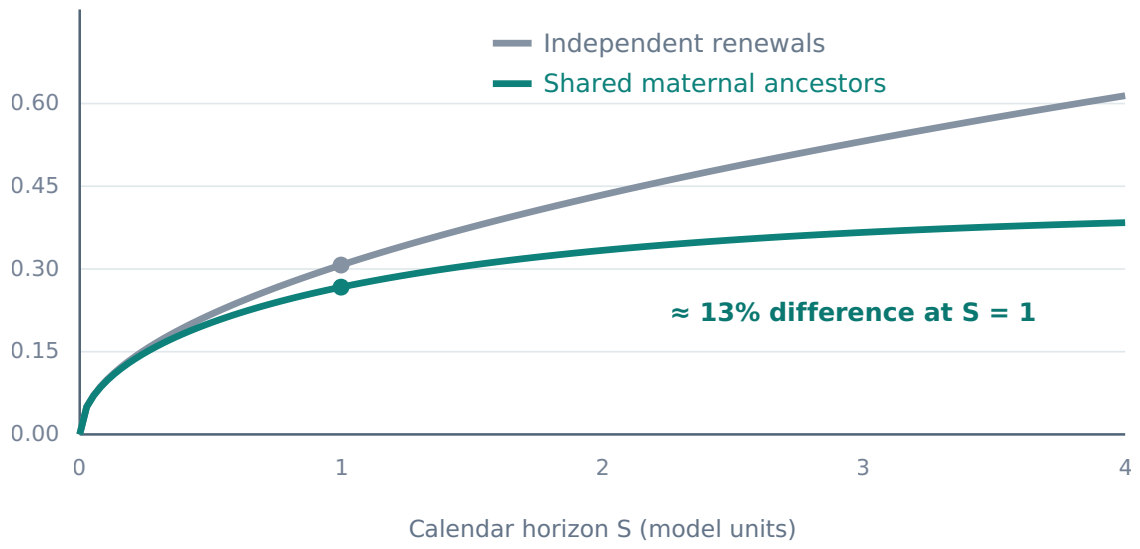
$$E[\Delta_N(t)] = E|G_1(t) - G_2(t)|$$

Combining this identity with the two-lineage limit and justified convergence of expectations yields **Theorem 9.4**:

$$\lim_{n\to\infty} E[\Delta_{Nn}(\lfloor nS \rfloor)] \; / \; \sqrt{n} = \sqrt{[\nu p(1-p)/(2-p)]} \cdot \operatorname{erf}[\sqrt{S}/(\sqrt{\nu(2-p)})]$$

Here *erf* is the standard error function. The theorem gives a coefficient for an *expected* cohort statistic in a fully specified stochastic model; it does not establish the realized disparity of a historical population.

5. How to read the “13%” figure



Asymptotic coefficient for expected cohort mean pairwise depth gap after normalization by √n. p = 0.5, ν = 1. No number is a historical human estimate.

Horizontal axis: the horizon S, in abstract model-time units rather than BCE calendar years. **Vertical axis:** the theoretical limiting coefficient of the expected mean pairwise generation gap, divided by √n; it is not a raw number of human generations.

The grey curve treats maternal renewals as independent. The blue-green curve keeps the same marginal generation-age distribution but allows the two ancestral identities to merge. Once they share a mother, their earlier paths are common, and the model predicts a smaller expected gap.

0.3071	0.2671
Independent-renewal coefficient at S = 1, p = 0.5, ν = 1.	Shared-ancestor coefficient with exactly the same illustrative parameters.

> **Correct interpretation:** (0.3071 − 0.2671) / 0.3071 ≈ **13%**. This is a reduction *in this mathematical example*, not a universal law or a measured effect in humans.

The figure cannot be read as an age of Homo sapiens, converted directly to years of historical divergence, or used to infer the worldwide maximum. Its percentages depend on the parameters and horizon.

6. What about the two most extreme people?

The maximum of a cohort is much more sensitive to rare outcomes than the typical gap. Theorem 9.2 provides a simultaneous concentration bound across every lineage of the homogeneous cohort, implying a √**(n log n)** upper-scale bound for the range on a fixed macroscopic horizon.

$$\sup_{0\le t\le nS} R_{Nn}(t) = O_P(\sqrt{(n \log n)}).$$

This bound yields **neither the exact law of the maximum nor a number for all living humans**. Another result controls particular types of simultaneous multiple merger for a sufficiently slowly growing sample; it is not a universal coalescent theorem.

7. The structure of ancestral blocks

Shared ancestral identities divide descendants into **blocks** with weights w_j. But block sizes alone are not enough: the depths, or *marks*, assigned to different blocks may be correlated. Propositions 8.1–8.2 give exact identities involving these covariances.

$$E[D] = 2dq(1-q)(1-\Sigma w_j^2) - 4d\,\Sigma_{i<j}\, w_i w_j\, \mathrm{Cov}(Z_i, Z_j).$$

Positive covariance can decrease expected dispersion, while negative covariance can increase it. That is why independent individual descriptions cannot settle a cohort-level question.

8. The larger M0–M6c project and its limits

The technical volume addresses renewal kernels, varying demographic environments, geography, reproductive weights, ancestral partitions, dense-to-sparse approximations and the compilation of historical parameters.

M6a–M6c specify a calibration architecture and its documentary obligations; they do not establish that historical parameters have been correctly measured for humans. Transferring structural marks from M4–M5 to genuinely observed mother–child generation counts requires a uniform approximation not established in full historical generality.

> **Scientific boundary:** none of these formulas gives a verified interval for mean or extreme maternal-generation differences among all living humans since the emergence of Homo sapiens. Sufficient novelty for a specialist journal is also yet to be independently assessed.

9. Where to verify every claim

This article is intentionally a doorway. The explicit-model derivations remain in the **24-page mathematical V0.6 manuscript**, while the **1,406-page M0–M6c corpus** preserves the larger programme and its documented limits.

Question	Proof in V0.6	Wider corpus
First maternal collision	Theorem 4.1, pp. 7–8	M3c, around p. 233
Time-varying environments / Cox	Theorem 5.1, pp. 9–10	M3, around p. 213
Expected cohort disparity	Theorem 9.4, p. 16	M4, around pp. 481–485
Uniform range control	Theorem 9.2, pp. 14–15	M4–M5, qualified claims
Ancestor-block marks	Propositions 8.1–8.2, p. 13	M4
Historical calibration	Section 11, pp. 19–20	M6, from p. 1035

The framework builds on prior renewal and coalescent theory (including Pyke, Kingman, Iglehart–Whitt, Sagitov–Jagers, Möhle and Wakeley). Independent assessment of the paper’s novelty is still required.

Conclusion

The scientific question is not only how many generations separate two people from the past, but how much of their maternal history is shared. Our model quantifies certain consequences of this dependence. Applying them to real human history remains a separate research challenge.

> Status: mathematical working preprint, not peer reviewed. No empirical historical-human estimates are claimed.